Hrithik Ravi

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EDUCATION

University of Wisconsin – Madison — *Madison, WI*

Incoming Ph.D. in Computer Sciences — *(Aug. 2026 –)*

University of Michigan — *Ann Arbor, MI*

M.S. in Elec. and Comp. Engineering (focus in Machine Learning, GPA: 4.0/4.0) — *(Aug. 2023 – Dec. 2024)*

B.S.E. in Computer Science and Engineering (Minor in Mathematics, Magna cum laude) — *(Aug. 2019 – April 2023)*

Honors: Dean's List/University Honors (Fall 2019, Fall 2022, Winter 2023)

PUBLICATIONS

H. Ravi, C. Scott, D. Soudry, Y. Wang. *The Implicit Bias of Gradient Descent on Separable Multiclass Data*. NeurIPS 2024.

RESEARCH EXPERIENCE

Q Labs (YC S23) – Research Intern — *San Francisco, CA*

Scalable Regularizers for Data-efficient Generalization – Advised by Prof. Andrew Gordon Wilson — *(April 2026 – present)*

- ❖ Searching for new regularizers beyond weight decay that will scale in the limited data, infinite compute regime.
- ❖ Performing multi-epoch pretraining with GPT-2 based language models on FineWeb.
- ❖ Implementing novel measures of complexity and compressibility; analyzing their correlation with/effect on validation loss.

University of Michigan AI Laboratory – Graduate Student Research Assistant — *Ann Arbor, MI*

Theoretical ML, LLMs – Advised by Profs. Ambuj Tewari, Rada Mihalcea, Clayton Scott — *(Aug, 2023 – April 2026)*

- ❖ Project 1: Designed synthetic reasoning task testbed to demonstrate brittle nature of existing post-training algorithms for frontier LLMs (GPT-4o). Implemented simple prompting scheme to surpass data-efficiency of supervised fine-tuning.
- ❖ Project 2: Researched cheap and interpretable algorithms to "steer" models away from biased activations. Observed negative results. Investigated and identified that observed bias was primarily an artifact of strict scoring metrics.
- ❖ Project 3: Mathematically proved that gradient descent directionally converges to simple solutions long after memorizing the training set. Published at NeurIPS '24.

Google DeepMind – Google Summer of Code — *Remote*

Benchmark for Multimodal LLMs – Advised by Paige Bailey — *(May 2025)*

- ❖ Designed research proposal and implementation plan for an open-source needle-in-a-haystack benchmark to probe multimodal long-context capabilities of Gemini.
- ❖ Proposal accepted to Google DeepMind (Google Summer of Code, 5% acceptance rate).

UofM Transportation Research Institute – Temp. ML Research Engineer — *Ann Arbor, MI*

3D Object Detection for Self-Driving Cars – Advised by Dr. Arpan Kusari — *(May 2023, June – Aug 2023: part-time)*

- ❖ Implemented a computer vision 3D IOU algorithm in OpenCV to evaluate performance of Detectron2 object detection model.
- ❖ Designed the algorithm to penalize false negatives harshly resulting in a model optimized for maximal passenger safety.
- ❖ Implemented a sensor fusion algorithm in MATLAB to provide a 360° view and enable safe path-planning in self-driving cars.
- ❖ Used Hungarian algorithm to fuse LIDAR and camera tracks, programmatically adding new tracks and discarding stale ones.

WORK EXPERIENCE

University of Michigan – Computer Science Graduate Student Instructor (TA) — *Ann Arbor, MI*

EECS 281: Data Structures and Algorithms — *(Aug. 2023 – Dec. 2024)*

- ❖ Received full tuition waiver to teach undergraduates foundations of data structures and algorithms.
- ❖ Held lectures, office hours to walk through example problems and to help students implement and debug C++ project solutions.

Amazon Web Services – Software Engineering Intern — *Herndon, VA*

Security and Networking; Time-series Analysis for Predictive Managed Scaling — *(June 2023 – Sept. 2023)*

- ❖ Built an iptables firewall with a token-bucket algorithm to fortify DNS servers against request floods, ensuring QoS for 3MN clients per region.
- ❖ Leveraged AWS Python API to implement an ML time-series pipeline in AWS Forecast, S3 to predict city-wide electricity consumption.

Amazon Web Services – Software Engineering Intern — *Austin, TX*

Full-Stack Mobile and Web Development for Supply Chain Analytics — *(May 2022 – Aug. 2022)*

- ❖ Mobile: Built notification UI in React Native, Typescript; backend in Expo, Firebase, Apple Notification Service, and NodeJS.
- ❖ Web: Developed interface to configure analytics, implementing frontend in React, backend in Kotlin and AWS DynamoDB.

INVITED TALKS

"A Theoretical Overview of Test-time Scaling", STATS 700 Guest Lecture (invited by Prof. Ambuj Tewari). *Sept. 2025*

"Is AI Changing Who We Are?", Michigan Friday Night AI, Invited Panelist (along with Prof. Rada Mihalcea and Prof. Jonathan R. Brennan). *Sept. 2025*

"Chain-of-Thought Reasoning: Trends and Directions", UMich Language and Information Technologies Lab. *May 2024*

"Multiclass Implicit Regularization", Michigan Student Symposium for Interdisciplinary Statistical Sciences (MSSISS). 3rd place. *March 2024*